

Multi-Source Candidate Data Transformer

Objective

Build a pipeline that ingests candidate data from multiple sources and generates one canonical candidate profile.

Sources

- Structured: Recruiter CSV
- Unstructured: Resume PDF

Pipeline

CSV + Resume → Extract → Normalize → Merge & Deduplicate → Confidence & Provenance →
Config Projection → Validate → Export JSON

Canonical Schema

- candidate_id
- full_name
- emails[]
- phones[]
- current_company
- title
- skills[]
- provenance
- overall_confidence

Normalization

- Phone → E.164 format
- Skills → Canonical names
- Missing values → null
- Remove duplicates

Merge Strategy

- Combine CSV and PDF data
- Deduplicate emails and phones
- Prefer normalized values
- Track source provenance

Features Implemented

- ✓ Multi-source ingestion
- ✓ Extraction and normalization
- ✓ Canonical profile generation
- ✓ Confidence scoring
- ✓ Provenance tracking
- ✓ Runtime configurable output
- ✓ JSON export

Edge Cases

- Empty CSV
- Missing phone number
- Duplicate emails
- Malformed PDF